

Topics :

- Introduction to Data centers
- Types of Data Centers
- Data center Classifications
- Networks of Small companies
- Network devices
- Hub vs Switch vs Router

Computer Networks

Data Centers - Assignment 2

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Introduction

Data center is a specialized facility used to house computer systems such as **servers, storage devices, and networking equipment**



What's the use of Data centers

Designed to provide reliable **power, cooling, security, and connectivity** so that organizations can store, process, and manage digital data and applications continuously and safely.

Types of Data Centers

- **Enterprise Data Centers** – Owned and operated by a single organization.
- **Colocation Data Centers** – Shared facilities where companies rent space for their servers.
- **Cloud Data Centers** – Operated by cloud providers to deliver scalable computing services.
- **Edge Data Centers** – Smaller facilities located closer to end users to reduce latency.
- **Hyperscale Data Centers** – Very large facilities designed for massive cloud workloads.

Data Center Classifications

Data centers are commonly classified into **four tiers** based on infrastructure redundancy, availability, and fault tolerance

(T1) – Basic Data Center

(T2) – Redundant Components

(T3) – Concurrently Maintainable

(T4) – Fault Tolerant

Data Center Classifications

T1 – Basic Data Center

Single path for power and cooling.

No redundancy for critical components.

Susceptible to downtime during maintenance or failure.

Suitable for small organizations with non-critical workloads.

EXAMPLES :

- Server rack inside an IT room
- File storage
- Email servers for small teams

Data Center Classifications

T2-Redundant Components

Includes redundant components such as extra generators or UPS.
Still uses a single power and cooling distribution path.
Improved reliability compared to Tier I.
Suitable for small to medium businesses.

EXAMPLES

- Internal business applications
- Company databases
- Web and email services

Data Center Classifications

T3– Concurrently Maintainable

Multiple power and cooling paths.

Redundant systems allow maintenance without shutting down operations.

High availability (about 99.98% uptime).

Commonly used by enterprises and service providers

EXAMPLES

- Multiple independent power and cooling paths
- Redundant UPS systems and backup generators
- Maintenance can be performed without shutting down IT services

Data Center Classifications

T4– Fault Tolerant

Fully redundant systems with no single point of failure.

Can withstand equipment failure without service interruption.

Highest availability (about 99.995% uptime).

Designed for mission-critical systems such as banking or government platforms.

EXAMPLES

- National payment systems
- Emergency and security services

Data Centers in emergency



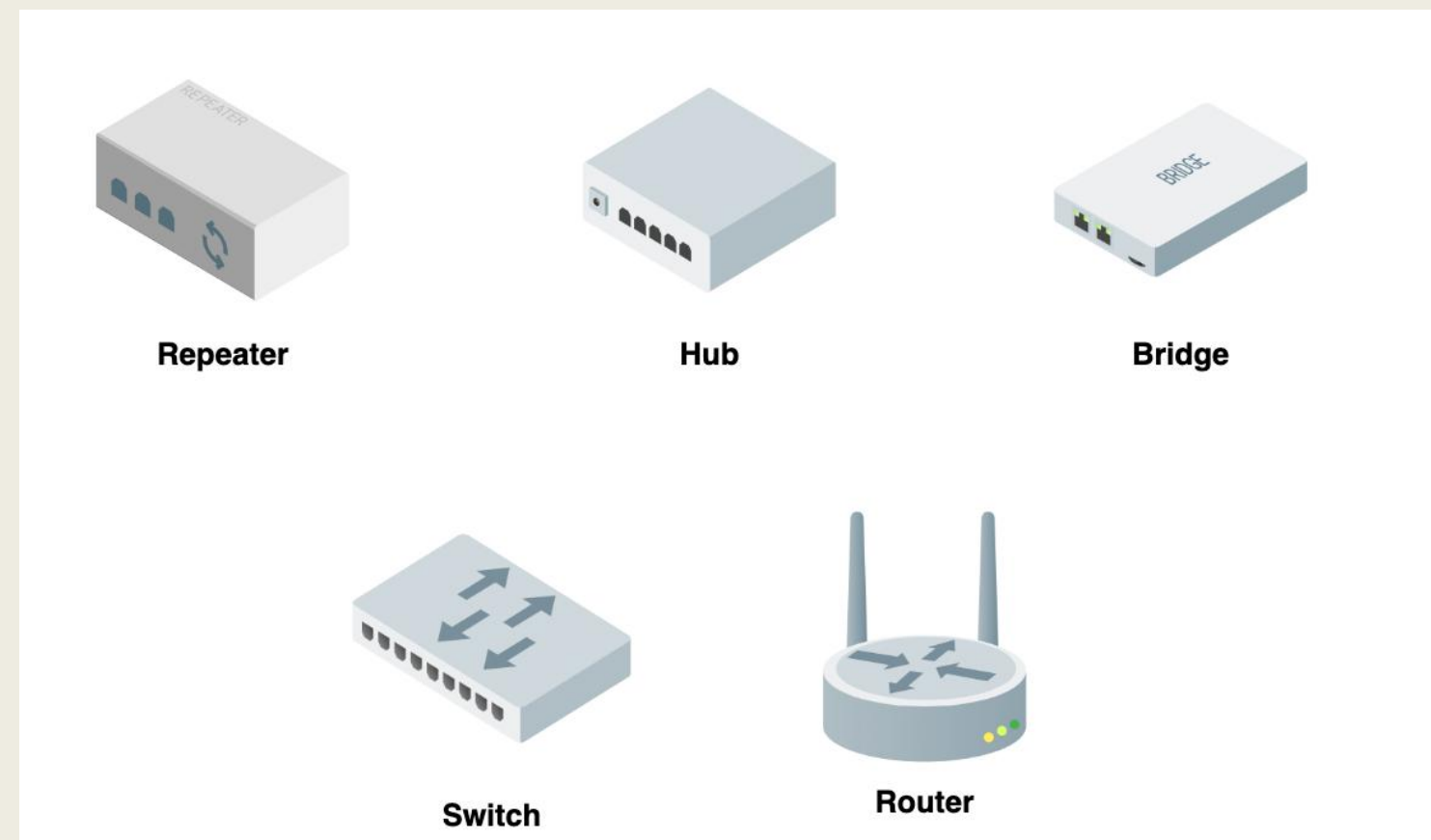
Data Center & Infrastructure Service Providers in Saudi Arabia

- **Sahara Net** – Operates data centers with colocation
- **NourNet** – A major ICT and data center provider offering colocation
- **STC Data Centers / Center3** – Part of Saudi Telecom Company's ecosystem, offering colocation
- **SSBS** – Offers Tier-level colocation, managed hosting, and ICT infrastructure services.
- **ICS Arabia** – Specializes in designing and building Tier III and Tier IV-ready data centers and enterprise infrastructure.
- **Solutions by STC** – A subsidiary of STC providing cloud, managed hosting, and enterprise digital infrastructure services.

Network Devices

Network devices are hardware components that connect computers and other digital equipment within a network. They allow devices to communicate with each other.

Some common examples include routers, switches, modems, and access points.



Network Devices

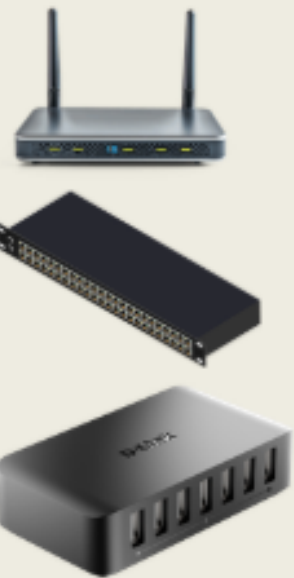
A Layer 3 device that forwards data between networks addresses and connects the LAN to the Internet.

A Layer 2 device that connects devices inside the local and forwards frames using MAC addresses.

Layer 1 device that broadcasts data to all ports without ; it is older and less efficient than a switch.

s Access Point (WAP): A device that provides wireless activity to the LAN, allowing Wi-Fi devices to access the

Layer	Device	Using :
Application		Protocols Services
Presentation		APIs
Session		
Transport	Hosts	TCP/IP
Network	Router	IP
Data Link	Switch	MAC
Physical	Hub	

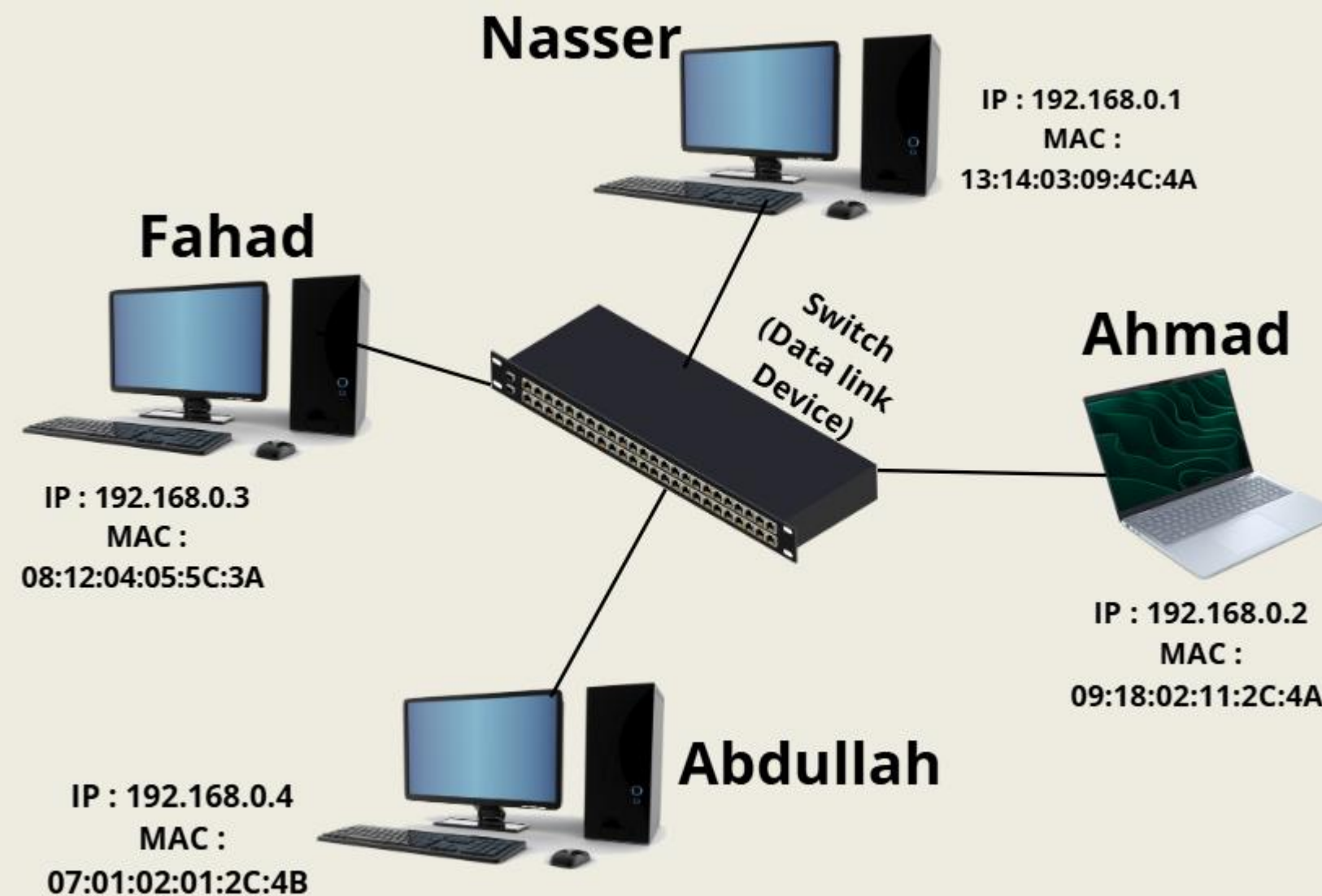


Overview of LAN – WAN

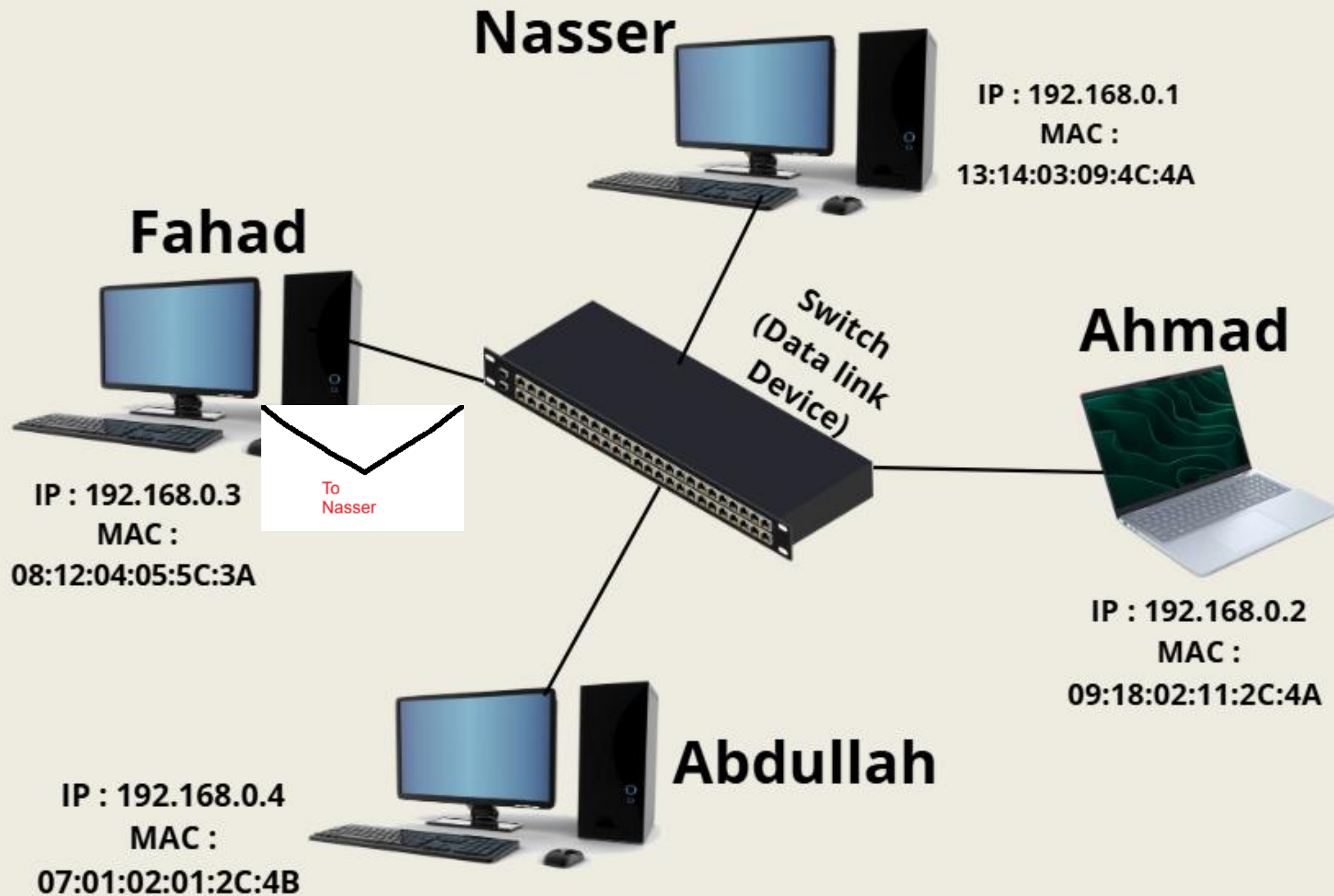
and using of devices

MAC (Medium Access Control) : or Physical Address. It's 6 Byte globally unique ID assigned by the manufacturer

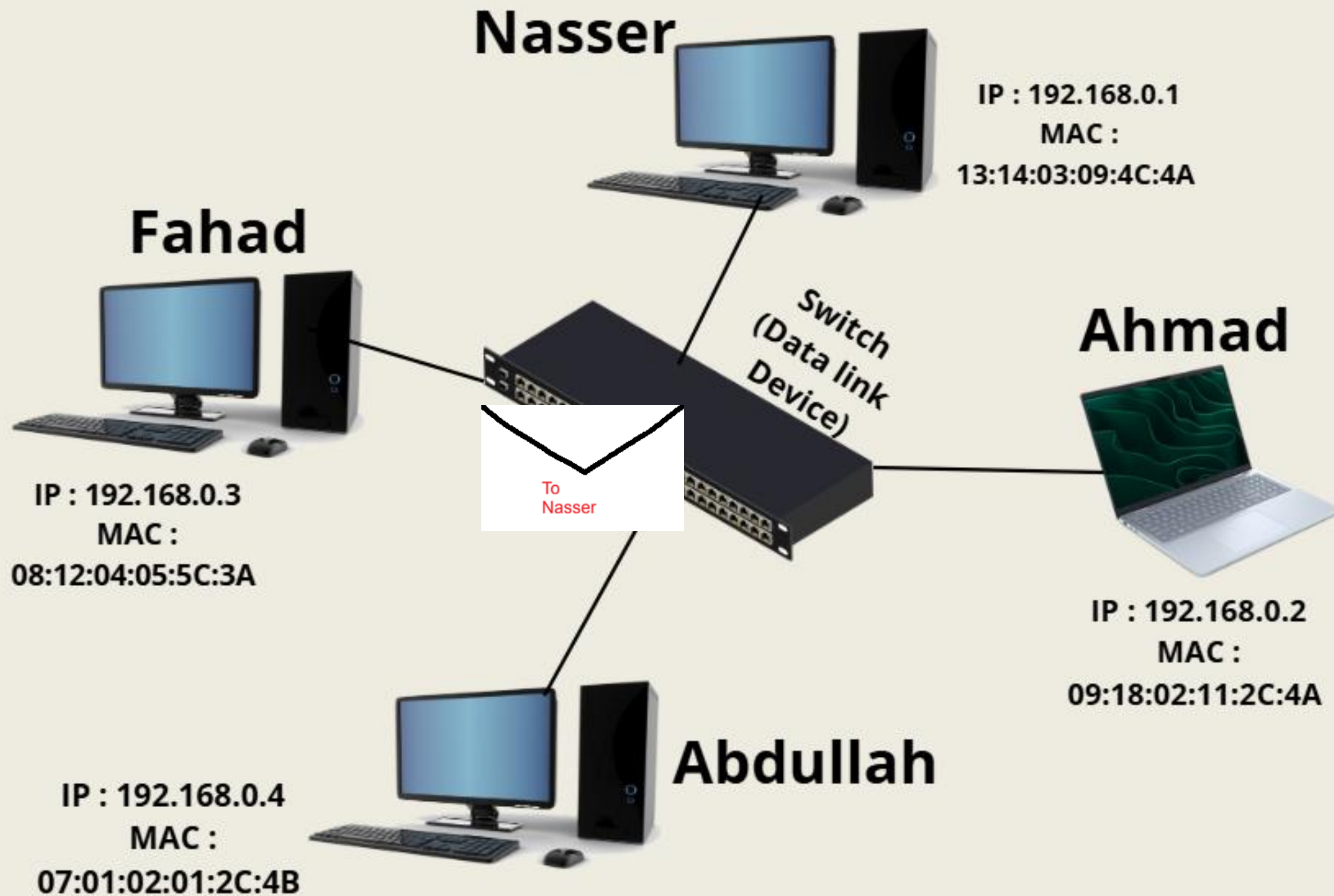
IP (Internet Protocol) : or Logical Address Used in network layer to access internet network.



Example



Example



Example

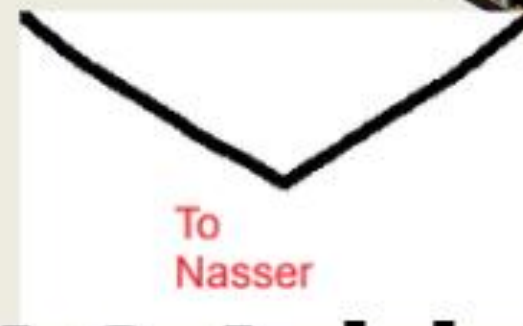
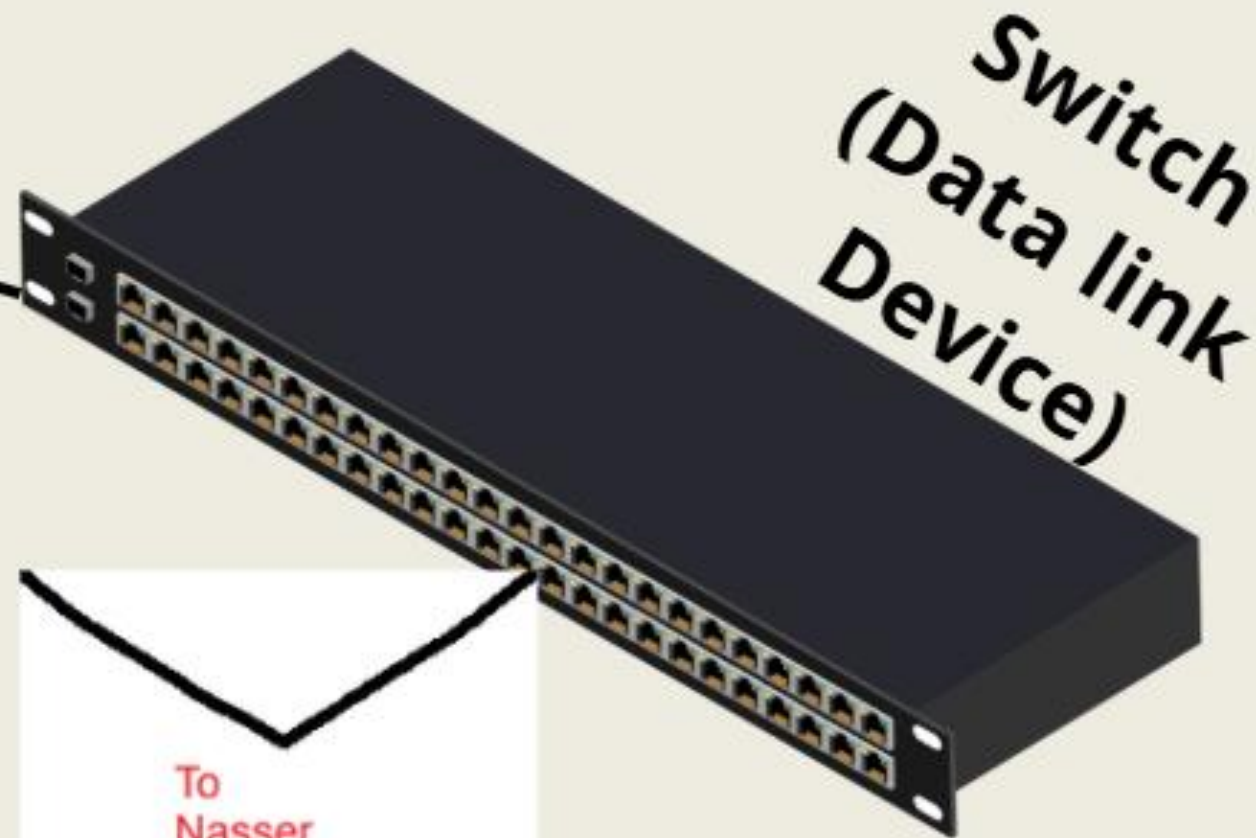
Fahad



192.168.0.3

MAC :

:04:05:5C:3A

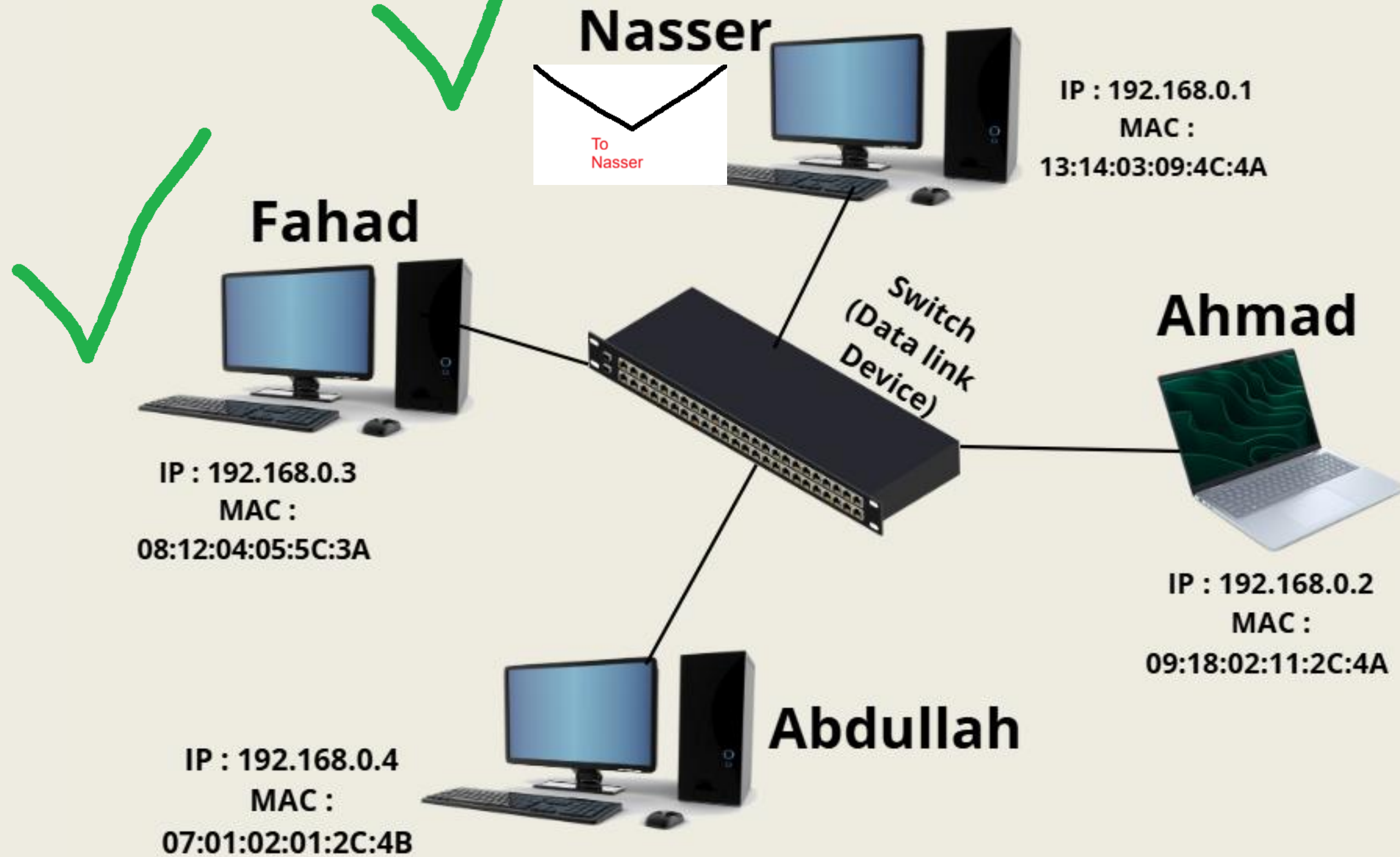


MAC Address

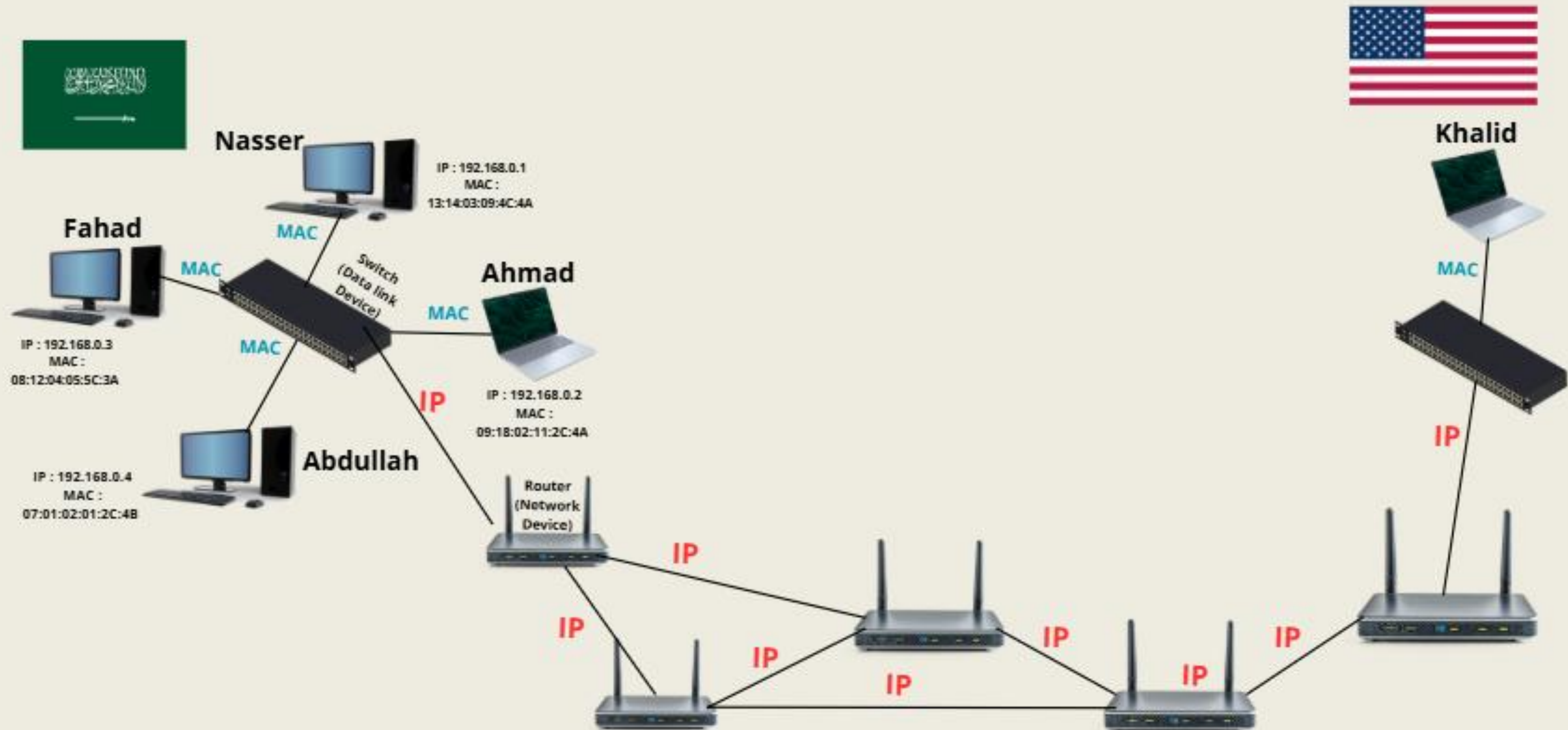
From 08:12:04:05:5C:3A Fahad

To 13:14:03:09:4C:4A Nasser

Example



From Switch to Router

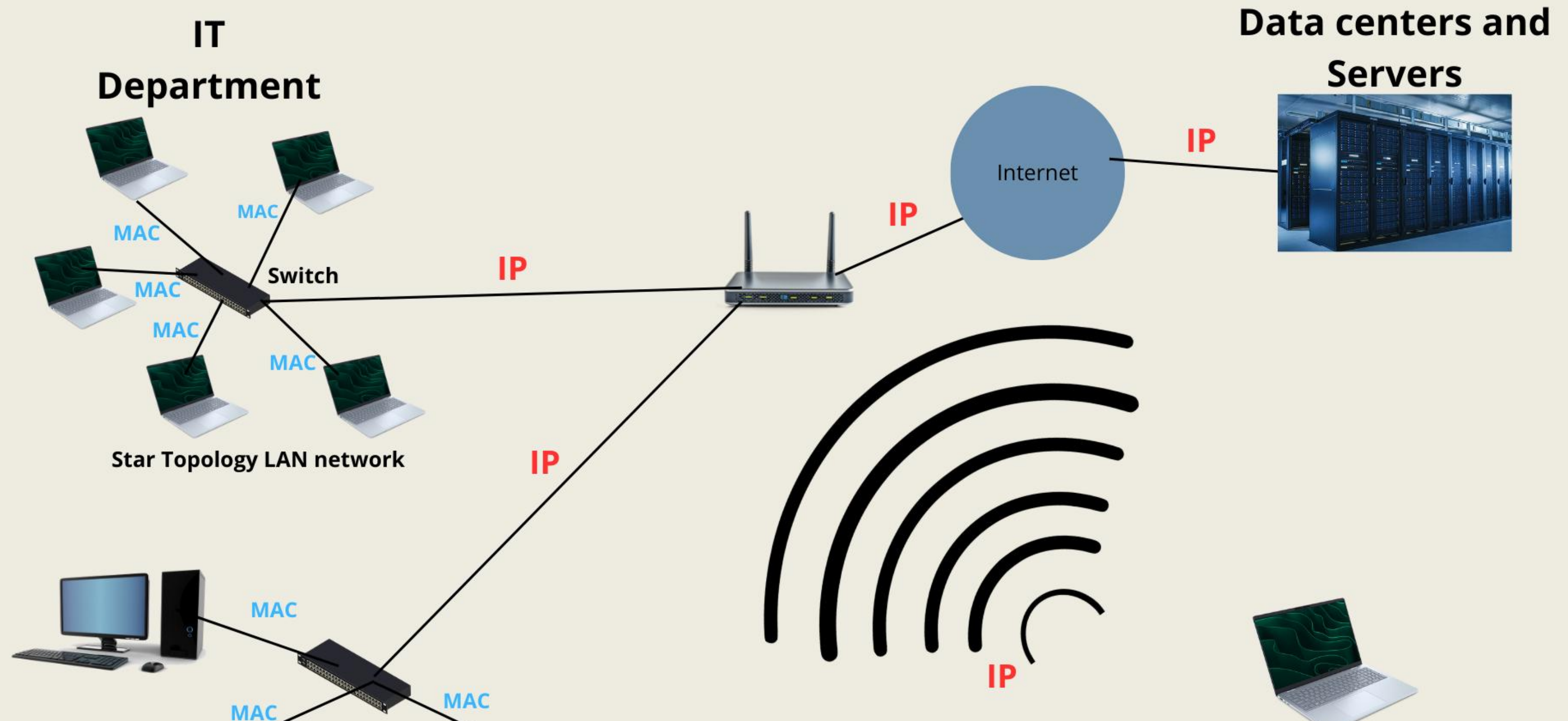


Networks in small companies

In small companies, computer networks play a vital role in supporting daily operations and communication , Most small businesses use simple network structures, such as Local Area Networks (LAN), which connect devices within the same office or building.

equipment like switches, routers, and wireless access points. Cloud services are also commonly used to reduce the need for expensive hardware.

Networks in small companies



Sources

- **Uptime Institute** – *Data Center Tier Classification System*
- **Cisco Systems** – *What Is a Data Center?*
- **NetworkChuck** – *Data center networking*
- **IBM** – *Data Center Overview and Services*
- **Amazon Web Services (AWS)** – *Types of Data Centers*

Thanks for listen